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4th Sem. / Auto, Mech (3rd /4th), Prod (3rd), T&D (3rd),GE, CNC, Adv. Manuf. Tech., Mecatronics, CAD/CAM,Mech Engg (Fabrication Tech.) Mech (CAD/CAM dsgn &Robotics)

Subject : Hydraulics and Pneumatics / Hyd. & Hyd. M/C

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 The mass density of water in S.I. units is

a) 1000 kg/m^3 b) 9810 kg/m^3

c) 9.81 N/m^3 d) 1000 kg/m^3

Q.2 Rain drops are spherical due to

a) Air resistance b) Surface tension

c) Viscosity d) Atmospheric pressure

Q.3 As a pressure head, the atmospheric pressure is

a) 1.033m of water b) 10.3 of water

c) 7.6m of water d) 76m of water

Q.4 The flow of liquid is said to be ______ when the path of individual particles do not cross one another.

a) Steady b) Unsteady

c) Laminar d) Turbulent

Q.5 In a reaction turbine, the water at inlet possesses

a) Only pressure energy

b) Only kinetic energy

c) Both pressure and kinetic energy

d) None of the above

Q.6 Centrifugal pump is started with its delivery valve kept ________

a) Fully open b) Fully closed

c) Partially open d) None of the above

Q.7 Valves in hydraulic system are used to control

a) Discharge of oil b) Pressure of oil

c) Direction of oil d) All of the above

Q.8 The inlet length of venturimeter is ______ the outlet length.

a) Equal to

b) More than

c) Less than

d) No relation between them

Q.9 The liquid used in inverted U-tube manometer should be of __________

a) Low density b) High density

c) High surface tension d) Low surface tension

Q.10 N/m is the unit of

a) Viscosity b) Capillarity

c) Surface tension d) Compressibility

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 The unit of discharge is ________.

Q.12 For a Laminar flow, the value of Reynold's number should be _________

Q.13 In Kaplan turbine, water flows to the axis of the shaft. (parallel/perpendicular)

Q.14 Hydraulic press is practical application of the _________ principle.

Q.15 Reciprocating pump is _________ displacement pump.

Q.16 Hydraulic actuator converts _________ into _____.

Q.17 The continuity equation shows _________ at two sections.

Q.18 The density of water is maximum at ____ degree C.

Q.19 Mercury is used for _________ pressure ranges.

Q.20 The co-effficient of venturimeter lies between _________.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Give the advantages and limitations of Manometers.

Q.22 Explain different types of flow.

Q.23 Distinguish between centrifugal pump and reciprocating pump.

Q.24 Explain various major losses.

Q.25 Give various industrial applications of Pneumatic hydraulic system.

Q.26 Derive relation between mass density and sp weight.

Q.27 Derive an expression for Pressure difference by inverted U-Tube.

Q.28 The diameters of an increasing diameter pipe at entry is 300mm and at exit 500mm. If the velocity at entry is 5m/s, then find the

i) discharge ii) velocity at exit

Q.29 Define critical velocity. Give its types and significance.

Q.30 If $3.7m^3$ of an oil weighs 34.25kN, find its specific weight, mass density and relative density.

Q.31 Derive an expression for loss of head due to friction.

Q.32 Write short note on hydraulic jack.

Q.33 Explain water hammer and its application.

Q.34 Draw a sketch of single acting cylinder and show its parts.

Q.35 What do you mean by priming? Name various methods for it.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 With the help of neat sketch, explain the construction and working of a Pelton wheel turbine.

Q.37 A pipe contains an oil of specific gravity 0.85. A differential manometer connected at the two points A and B of the pipe shows a difference in mercury level as 19cm. Find the difference of pressure at the two points.

Q.38 Give various common faults and their remedies in a Hydraulic system.